



Vision Language Models Know Law of Conservation without Understanding More-or-Less

Growing AI Like A Child
<https://growing-ai-like-a-child.github.io/>

Dezhi Luo, Haiyun Lyu, Qingying Gao, Haoran Sun, Yijiang Li, Hokin Deng
UMich, UNC, JHU, UCSD, CMU

Introduction

Conservation is a critical milestone of cognitive development considered to be supported by both the understanding of quantitative concepts and the reversibility of operations. To assess whether this critical component of human intelligence has emerged in Vision Language Models, we have curated the **ConserveBench**, a battery of 365 cognitive experiments across four dimensions of physical quantities: **volume, solid quantity, length, and number**. The former two involve transformational tasks which require reversibility understanding. The latter two involve non-transformational tasks which assess quantity understanding.

Surprisingly, we find that while Vision Language Models are **generally good at transformational tasks**, they tend to fail at **non-transformational tasks**. There is a dissociation between understanding the reversibility of operations and understanding of quantity, which both are believed to be the cornerstones of the understanding of law of conservation in humans.

Conservation Tasks

Conservation is the ability required for understanding that physical quantities do not change in amount when they undergo spatial transformations.

Following classic Piagetian design, we assessed four dimension of physical quantity: number, length, solid, and liquid. **Transformation** process is operationalized into three phase.

Additionally, **non-transformational** tasks consists of single-image tasks featuring the end phase of Number and Length dimensions to probe quantity understanding and its relationship with conservation.

A

The above three images represent a continuous process from time frame 1 to 3. Please answer, is the number of coins in the upper row and lower row in the third image the same? Please select from the following choices and provide an explanation.

A. Yes
B. No, the upper row has more coins
C. No, the lower row has more coins

Correct Answer: A

GPT4o: "To determine the number of coins in the upper and lower rows in the third image: in the third image, there are 5 coins in the upper row and 5 coins in the lower row, thus, the number of coins in the upper row and lower row in the third image is the same. **The correct choice is: "a yes"**

B

The above three images represent a continuous process from time frame 1 to 3. Please answer, in the third image, is the length of the lower straw longer than that of the upper straw? Please select from the following choices and provide an explanation.

A. Yes
B. No, they are the same
C. No, it's the other way around

Correct Answer: B

GPT4o: "To determine the length of the straws in the third image: in the first image, both straws appear to be the same length, in the second image, the straws are adjusted, but it seems to change the perceived alignment, not the physical length; in the third image, both straws look the same length if laid out in parallel without any overlap or perspective distortion, thus, the length of the upper straw and the lower straw in the third image seems to be the same. **The answer is "b no, they are the same"**

C

The above three images represent a continuous process from time frame 1 to 3. Please answer, is the size of the play dough in the first image bigger than the size of the play dough in the third image? Please answer with Yes or No and provide an explanation.

Correct Answer: No

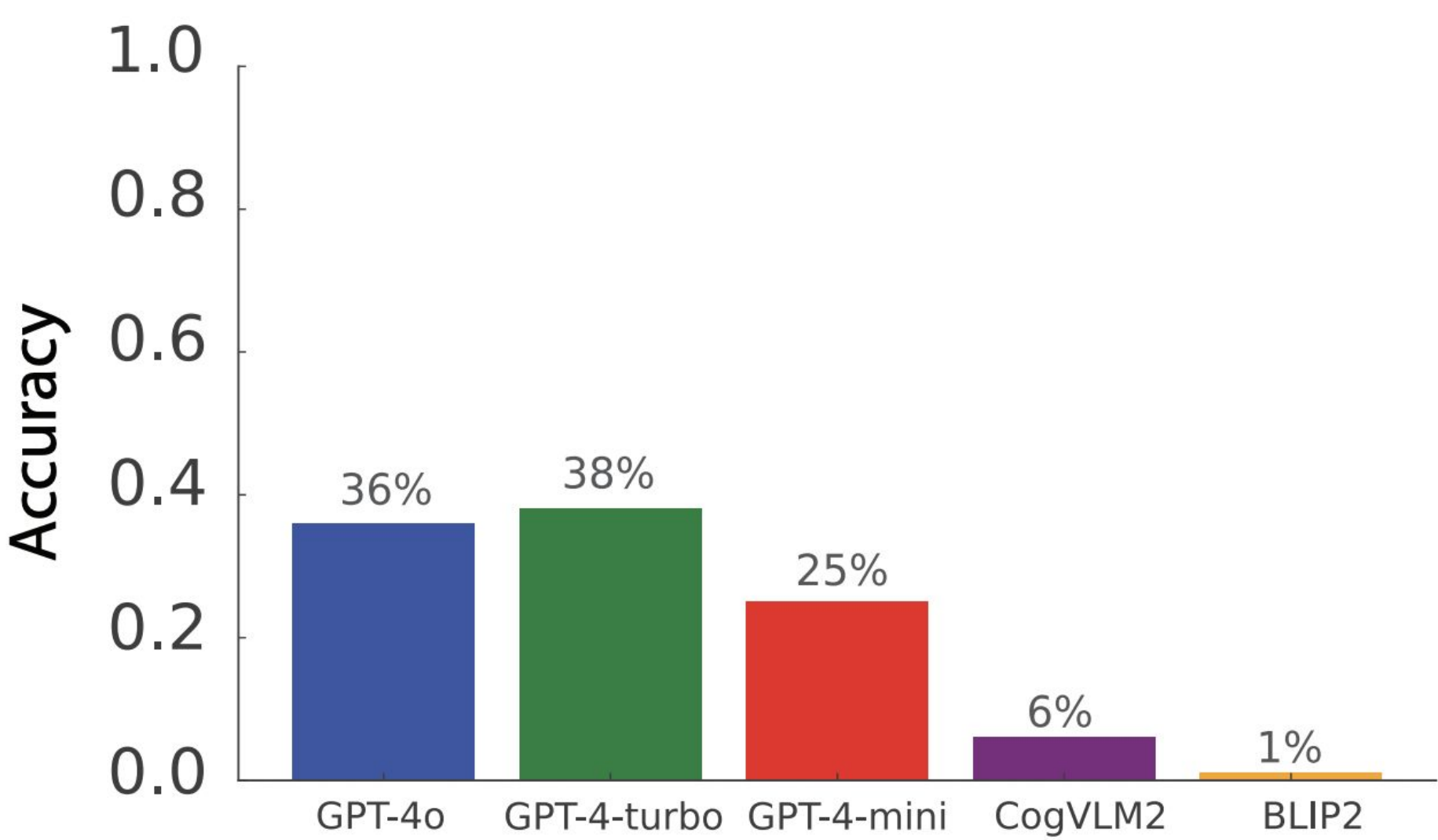
GPT4o: "No, the size of the play dough in the first image is not bigger than the size of the play dough in the third image. The play dough appears to have been reshaped from a compact form to a flattened form while the shape and apparent volume may seem different due to the change in form, the actual amount of play dough remains the same"

D

The above three images represent a continuous process from time frame 1 to 3. Please answer, does the glass on the left in the first image contain more liquid than the glass on the right in the third image? Please answer with Yes or No and provide an explanation.

Correct Answer: No

GPT4o: **Yes** "Yes, the glass on the left in the first image contains more liquid than the glass on the right in the third image the process depicted **shows the liquid being poured from the glass to the bowl**, which indicates that the amount originally in the glass is more than what ends up in the bowl, **considering potential spillage or absorption**"



Dissociation Between Conservation and Simple Quantity Assessment

- VLMs perform well on Transformational Tasks, possibly suggesting ability to recognize the reversibility of physical operations.
- However, in Non-Transformational Tasks, particularly for number and length dimensions, VLMs perform significantly worse, making consistent errors similar to pre-operational children with limited understanding of quantity. This indicates that VLMs struggle with tasks that require a basic conceptual understanding of quantity.

Interpretation: conservation helps to reject unreliable quantity judgment heuristics

- The heuristics that visually longer or bigger stuff tend to have more stuff among them remain presented 'in older children and adults as a common System-1 strategy used in daily life
- But can be suppressed with effortful, System-2 thinking strategy

